Performance Testing

Date	15 March 2026
Team ID	SWTID-2026-1940
Project Name	Team Enrollment Confirmation - EduGenie: Google Gemini Powered Learning Assistant
Team Members	R. Priya dharshini, S. Shobana, S. Rakshana, E. Shylaja, S. Sheeba
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the EduGenie learning assistant behaves under expected and peak load conditions. The report documents the testing approach, scenarios, performance indicators, observations, and team responsibilities.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter / Locust
Type of Testing	Load Testing, Stress Testing, and Spike Testing
Target Module / API	EduGenie FastAPI backend APIs, Gemini-powered Q&A; and learning-assistance endpoints
Test Environment	Local / Hosted Server environment with application backend and Google Gemini API integration
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Baseline Load Test	50	300	Stable response time and no significant errors.
2	Peak Load Test	200	300	Average response time remains below target and error rate stays below 1%.
3	Stress Test	250	180	System degrades gracefully without crashes; resources remain manageable.
4	Spike Test	200 → 50	120	System recovers after a sudden traffic increase and decrease.

Reference note: The documented project performance baseline reports an average response time of 1.45 seconds under a peak load of 200 concurrent virtual users with a 0.25% error rate. These figures should be replaced with the team's own exported test results if a fresh run was performed.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.45 seconds	Pass	Within documented target.
2	Response Time (Max)	< 5 seconds	Not documented	—	Capture max latency from the final test report.
3	Throughput (Req/sec)	Project-specific	Not documented	—	Record requests/sec from JMeter/Locust output.
4	Error Rate	< 1%	0.25%	Pass	Below target in documented peak-load result.
5	CPU Utilization	< 80%	Not documented	—	Capture server CPU utilization during peak load.
6	Memory Utilization	< 80%	Not documented	—	Capture server memory utilization during peak load.

Step 4: Observations & Analysis

Key Findings:
The documented peak-load result indicates that the application maintained an average response time of 1.45 seconds at 200 concurrent virtual users, with a 0.25% error rate. The available project documentation also describes caching and asynchronous processing as performance-oriented optimizations.
Bottlenecks Identified:
PDF/document analysis can be CPU-intensive. The project documentation identifies document parsing as a bottleneck area and describes asynchronous task processing as an optimization. Gemini API latency and external-service availability can also affect end-to-end response time.
Optimization Steps Taken:
The project documentation describes asynchronous task queues for CPU-heavy document parsing and Redis caching for frequently queried concepts to reduce repeated processing and Gemini API latency/token usage.

Step 5: Screenshots / Evidence

Evidence 1: Insert JMeter/Locust summary showing users, response time, throughput, and errors.
Evidence 2: Insert CPU/memory monitoring graph or server metrics captured during the peak-load test.

Source for the documented baseline and architecture reference: project repository documentation for “EduGenie: Google Gemini-Powered Learning Assistant.”

Team Work Allocation

Responsibilities are assigned in the same order as the names provided. Each member can attach evidence such as code commits, screenshots, test files, reports, or presentation material.

S.No	Team Member	Assigned Work / Responsibility	Expected Evidence
1	R. Priya dharshini	**Team Lead & Performance Testing Coordinator:** coordinate testing plan, define load/stress/spike scenarios, execute or supervise JMeter/Locust runs, consolidate results, and prepare the final performance report.	Test plan, JMeter/Locust report, consolidated PDF, test summary.
2	S. Shobana	**Backend & API Testing:** identify target FastAPI endpoints, prepare API test data, validate response behavior under concurrent requests, and document API-level observations.	API test cases, endpoint screenshots, response-time observations.
3	S. Rakshana	**AI/Gemini Integration Testing:** test Gemini-powered Q&A;, summarization, quiz, explanation, and learning-path interactions; record latency/error behavior related to AI calls.	AI test scenarios, sample requests/responses, latency/error evidence.
4	E. Shylaja	**Performance Monitoring & Optimization:** monitor CPU/memory usage, identify bottlenecks, review caching/asynchronous processing effects, and document optimization observations.	CPU/RAM graphs, bottleneck notes, optimization screenshots.
5	S. Sheeba	**Documentation, Evidence & Presentation:** organize screenshots and test evidence, maintain result tables, prepare observations/analysis, and support final demonstration/presentation.	Evidence folder, formatted report, presentation slides, demo checklist.

Final Submission Checklist

■	JMeter/Locust test plan and execution results attached.
■	Response time, throughput, error rate, CPU and memory metrics recorded.
■	Screenshots/evidence inserted into Step 5.
■	All team members' assigned work and evidence are documented.
■	Final report reviewed before submission.

Important: Values marked “Not documented” are intentionally left open rather than invented. Replace them with the actual measurements from your team's JMeter/Locust run before final submission.